BATCH & BUILD · STEM LEARNING SHEET · AGES 8–12

Volcano: Fizz, Foam, and Eruptions

THE SCIENCE

Baking soda is a base, and vinegar is an acid. When they mix, they react and make a new gas, carbon dioxide. The gas bubbles want space, so they push the liquid up and out of the volcano's opening. Dish soap traps the bubbles and makes thicker foam that looks like lava.

Real volcanoes work differently. Deep underground, rock gets so hot it melts into magma, which has gases trapped inside, a bit like a shaken soda. When pressure builds, the magma pushes up through cracks and erupts. Once magma reaches the surface, it's called lava.

KEY WORDS

Acid: a substance like vinegar or lemon juice that tastes sour.

Base: a substance like baking soda that reacts with acids.

Chemical reaction: when substances mix and make something new.

Carbon dioxide: a gas, the same one you breathe out.

Magma: melted rock underground.

Lava: melted rock that has reached the surface.

Pressure: a pushing force.

EXPERIMENT – THE BIGGEST ERUPTION

Question: Does warm vinegar make a bigger eruption than cold vinegar? Every trial uses 1 tablespoon of baking soda and 1/4 cup of plain vinegar. Only the temperature changes.

1. Cold: 1/4 cup of vinegar straight from the fridge.
2. Room temperature: 1/4 cup of vinegar left on the counter for an hour.
3. Warm: 1/4 cup of vinegar warmed by an adult (warm, not hot).

Measure how high the foam rises against a ruler taped behind the volcano. Do each trial 3 times and find the average.

Think about it: Why might heat speed up a reaction?

SAFETY

Wear safety glasses, keep vinegar away from eyes, and do it on a tray or outside.

CHECK YOUR UNDERSTANDING

1. What gas do baking soda and vinegar make?
2. Is vinegar an acid or a base?
3. What's the difference between magma and lava?
4. What does dish soap do in the eruption?
5. What builds up inside a real volcano before it erupts?

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ANSWER KEY (FOR GROWN-UPS)

1. Carbon dioxide.
2. Acid.
3. Magma is underground; lava is on the surface.
4. Traps bubbles to make thicker foam.
5. Pressure from gas and magma.

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